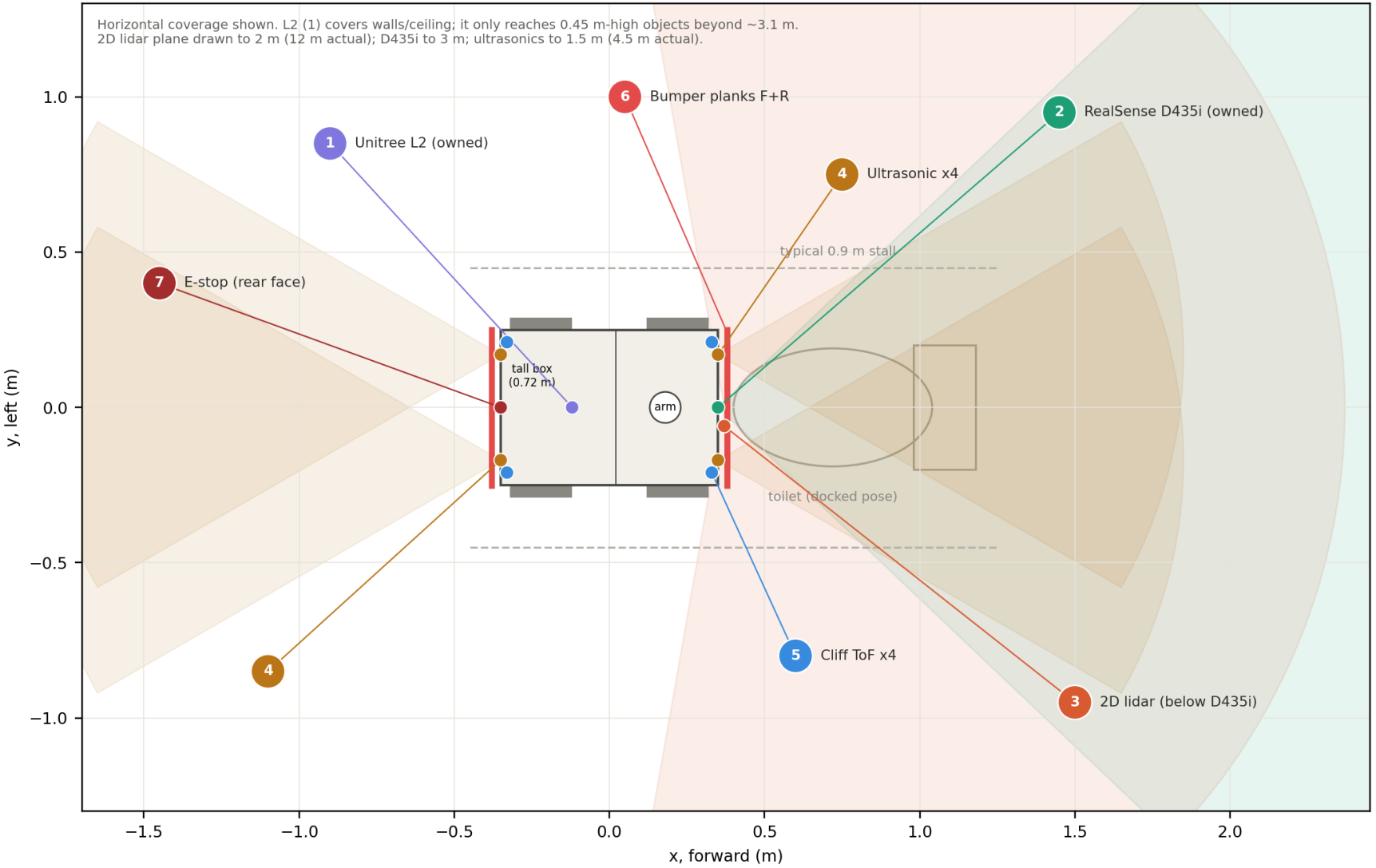
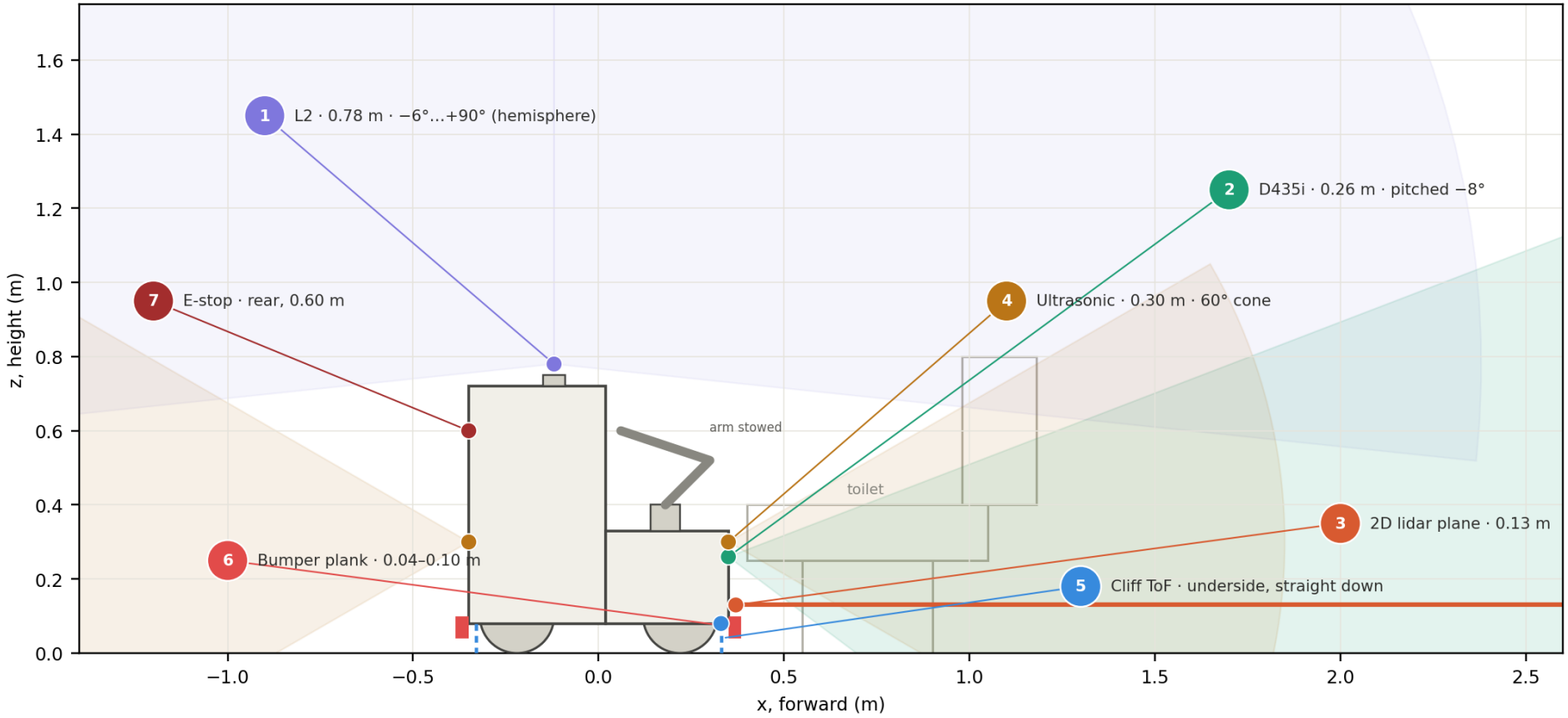
Sensor placement · top view

Frame: origin at robot centre on the floor, x forward (toward the toilet), y left, z up. Body assumed 0.70 × 0.50 × 0.72 m with a 0.33 m front deck, scaled from the CAD renders. Owned: Unitree L2, RealSense D435i, Jetson Orin Nano Super.

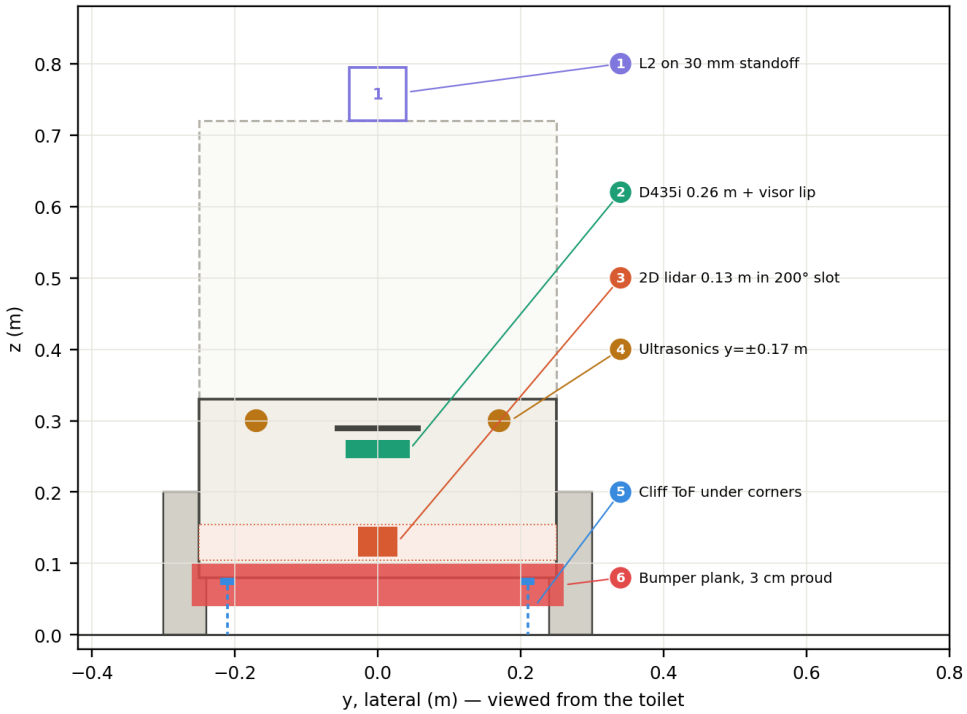


Sensor placement · side view

Vertical coverage of forward- and rear-facing sensors along the centreline, docked at the toilet.



Sensor placement · front face and mounting schedule



#	Sensor	Position x, y, z (m)	Orientation and mounting notes
1	Unitree L2 (owned)	−0.12, 0, 0.78	Box top on 30 mm standoff, upright, negative-angle mode. No window over the optics; out of splash zone.
2	RealSense D435i (owned)	0.35, 0, 0.26	Front face centre, facing +x, pitched −8°. Printed visor lip above; no window (IR projector back-reflects).
3	2D lidar	0.37, 0, 0.13	Front face centre in a 200° horizontal slot, 2 cm proud of the face.
4	Ultrasonic ×4	±0.35, ±0.17, 0.30	Front pair facing +x, rear pair facing −x. Rear pair is the main reverse sensor.
5	Cliff ToF ×4	±0.33, ±0.21, 0.08	Underside corners ahead of each wheel, pointing straight down. One XSHUT line each.
6	Bumper planks	front and rear, z 0.04–0.10	Floating planks on microswitches, 3 cm proud of the face so they hit before the lidar.
7	E-stop	−0.35, 0, 0.60	Rear face of the tall box, reachable from behind and both sides.
—	Internal	inside body	Orin Nano, ESP32 hub, drive relay, DC-DC converters, RC receiver (antenna clear of metal).

Bill of materials · additional parts

All linked vendors are Indian online retailers that ship within India. Prices include GST unless noted. Stock is as shown on the listing when checked; “est.” marks prices from memory, not a listing.

#	Item	Qty	Primary choice (brand · Indian vendor · price · stock)	Alternative (different brand where available)	Est. total ₹
3	**2D lidar**	1	SLAMTEC RPLIDAR C1 (C1M1-R2) · Robu · ₹6,749 · in stock	LDROBOT LD19 DTOF lidar (UART) · Robu · out of stock, 25–30 day restock	6,749
4	**Waterproof ultrasonic**	4	DFRobot A02YYUW (IP67, 3 cm blind zone, UART) · Robu · ₹1,698 each · in stock. Also Thingbits at ₹1,698 (out of stock)	JSN-SR04T v3.0 (generic) · Zbotic · ~₹200–350 each. 20–25 cm blind zone; control board not waterproof	6,792
5	**Cliff ToF sensor**	4	VL53L1X ToF module · Probots · price on listing (est. ₹500–700)	SparkFun VL53L4CD Qwiic · Tanotis · ₹3,000 each incl. duty. Both share I2C address 0x29: wire one XSHUT per sensor	2,400 est.
6	**Bumper microswitches + planks**	4–6	Lever microswitch with NO + NC (SPDT), plus plank and return springs · commodity part, any vendor	No specific listing verified	500 est.
7	**E-stop, 22 mm latching**	1	Schneider Harmony XB5AS8445 (1NO + 1NC, IP66/67) · Schneider India eShop · ₹2,653 excl. tax, stock unclear. Also LionCircuits · ₹4,296 · ships in 7–10 days	Generic XB2-BS142-style latching mushroom (~₹300–600) · no specific listing verified	3,130
8	**Drive cut-off relay + flyback diode**	1	SLA-12VDC-SL-A, 12 V coil, 30 A @ 30 VDC · Robu · ₹69 · in stock	Songle SLA-12VDC-SL-C (SPDT) · Robocraze. Only for a drive bus ≤ 30 V DC; above that use a rated DC contactor	200 est.
9	**RC kill switch (TX + RX with failsafe)**	1	FlySky FS-i6X + FS-iA10B · Robu · price not shown on listing; buyer reviews cite under ₹6,000. Plus an RC PWM switch module to drive the kill relay (commodity)	No alternative brand verified in an Indian store	6,300 est.
10	**DC-DC buck converters**	2 + 1	XL4016 (XH-M401), 8–36 V in, adjustable out · Robu · price not shown. One set to 19 V (Jetson), one to 12 V (L2); a small 5 V buck for MCU and sensors	Mean Well DDR-60G-12 (12 V, 5 A, 9–36 V in) · Indian listing not verified (~US$33 abroad)	1,200 est.
11	**Sensor hub MCU**	1	Espressif ESP32-DevKitC (WROOM-32E) · Robu · ₹799 · in stock	STM32 Nucleo or Raspberry Pi Pico 2 · no specific listing verified	799
12	**NVMe SSD, M.2 2280 (logging)**	1	Crucial P3 · MDComputers · 1 TB listed at ₹4,650, shown out of stock	Samsung 980 250 GB or Crucial P3 500 GB · Microcenter India NVMe range	3,500 est.
13	**Power distribution**	1 set	Inline fuses + holders, XT60 pairs, 14/18 AWG silicone wire, 1N5408 diode · commodity	—	1,000 est.
14	**PETG filament**	1 kg	For mounts, D435i visor, lidar slot shroud · commodity	—	1,200 est.
	Total, additional parts				**≈ ₹34,000**

Item numbers 3–7 match the drawing callouts; 1–2 are the owned L2 and D435i. The XL4016 needs a battery bus of roughly 22–36 V to produce 19 V; with a 12 V battery use a boost converter instead. Cliff sensors were switched from VL53L4CD to VL53L1X on cost: Indian VL53L4CD breakouts list at about ₹3,000 each. Before ordering, re-check stock on the Robu listings marked “price not shown”.